
INCONSISTENCY IS MOTION: A FLOW-FREE DETECTOR FOR DYNAMIC 3D RECONSTRUCTION

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ABSTRACT

Feed-forward pointmap regression has become the dominant route to dense multi-view reconstruction: a network maps each image pair to per-pixel 3D coordinates in a shared frame, so that depth, relative pose, and dense correspondence emerge together. Turning many such pairwise predictions into a single scene still requires a *fusion* step, handled today either by a per-scene global-alignment optimization or by a learned recurrent state. Neither makes explicit *where*, and to *what degree*, the pairwise predictions fail to admit a globally consistent reconstruction—a question that is a nuisance for rigid scenes but central for dynamic ones, where object motion makes a single consistent rigid geometry genuinely obstructed rather than merely hard to find. We recast multi-view pointmap fusion as the search for a *global section* of a cellular sheaf on the view graph, whose stalks carry per-view geometry and whose restriction maps carry the network’s predicted relative transforms. In this language a globally consistent reconstruction is the kernel of the sheaf Laplacian, recoverable by a sparse, graph-native diffusion with a spectrally controlled rate, while the obstruction to consistency is measured by the first cohomology, whose mass localizes on exactly the moving and occluded regions. The formulation subsumes the correspondence graph as a strictly less expressive special case, yields a principled alternative to the opaque global-alignment optimizer, and produces an unsupervised dynamic-region detector as a by-product. This document develops the formulation and the preliminaries it rests on.

1 Introduction

Pointmap regression has reshaped multi-view 3D reconstruction. In place of the classical pipeline—detect correspondences, estimate poses, triangulate, fuse—models in the DUST3R family [Wang et al., 2024, Leroy et al., 2024] learn a single feed-forward map from an image pair (I^1, I^2) to two dense *pointmaps*, per-pixel 3D coordinates expressed in a common camera frame, so that depth, relative pose, and dense correspondence are read off a single prediction. For dynamic scenes, where object motion breaks the rigid-alignment assumption, MonST3R [Zhang et al., 2025] retrains on moving content and D²UST3R [Han et al., 2026] introduces static–dynamic aligned pointmaps that warp static pixels by camera pose and dynamic pixels by optical flow, sharpening correspondence on moving objects.

These models are inherently *pairwise*. Recovering a scene from $N > 2$ views requires a separate *fusion* stage that reconciles the overlapping pairwise predictions into one globally consistent geometry. Two families address it. The first keeps a per-scene *global-alignment* optimization: the original DUST3R formulation minimizes pairwise pointmap disagreement over per-view depths, poses, and intrinsics by gradient descent for a few hundred iterations [Wang et al., 2024], an objective inherited with minor variation by its dynamic successors. The second replaces optimization with a learned state: Spann3R carries an external spatial memory [Wang and Agapito, 2025], CUT3R maintains a persistent recurrent state that emits each new pointmap directly in a shared frame [Wang et al., 2025], and SLAM-oriented systems fuse submaps with loop closures [Murai et al., 2025].

Across both families, global consistency is enforced *implicitly*. The optimizer drives pairwise residuals toward zero; the learned state absorbs multi-view agreement into its weights. Neither exposes the object that actually controls

whether fusion can succeed: a *principled, computable account of the obstruction* to gluing the local predictions into one scene. For rigid scenes this absence is tolerable, because a consistent global geometry exists and the only question is how fast one finds it. For dynamic scenes it is the crux. A moving object photographed at two times has *no* single rigid placement consistent with both views; the pairwise predictions are locally correct yet globally incompatible, and the incompatibility is not noise to be averaged away but structure to be measured. Current fusion either fights this obstruction (rigid alignment, which smears the error into neighboring static geometry) or hides it (a learned state, which offers no handle on where consistency genuinely failed).

Our viewpoint. The pairwise predictions carry exact algebraic structure that names this obstruction. Index the views by the vertices of a *view graph* $G = (V, E)$, with an edge for each pair the network has related. Each view supplies a local description of the scene’s geometry; each edge supplies a predicted transform relating the two descriptions on their overlap. A globally consistent reconstruction is precisely an assignment of a geometry to every view that agrees, edge by edge, under these transforms—in the language of applied topology, a *global section* of a cellular sheaf [Curry, 2014, Ghrist, 2014] whose stalks are the per-view geometries and whose restriction maps are the predicted transforms. We make this identification precise and build the fusion problem on it. The sheaf is the canonical structure for integrating local observations into a global state [Robinson, 2017]; our contribution is to recognize that multi-view pointmap fusion is an instance, and that its dynamic-scene difficulties are exactly the sheaf’s cohomology.

What the viewpoint buys. Three consequences follow, and they organize the paper. (i) *Consistency is a kernel.* A reconstruction is globally consistent if and only if it lies in the kernel of the *sheaf Laplacian* $\mathbf{L}_{\mathcal{F}}$, a sparse positive-semidefinite operator built from the restriction maps [Hansen and Ghrist, 2019]. Fusion becomes a graph-native diffusion—heat flow under $\mathbf{L}_{\mathcal{F}}$, a sequence of sparse linear solves—whose convergence rate is governed by the spectral gap of $\mathbf{L}_{\mathcal{F}}$, replacing the opaque gradient descent of global alignment with an operator one can analyze and precondition. (ii) *Inconsistency is cohomology, and it localizes.* When no global section exists, the failure is measured by the first cohomology $H^1(G; \mathcal{F})$, equivalently by the harmonic 1-cochains in the cokernel of the coboundary. In a dynamic scene this obstruction mass concentrates on the edges and regions where motion and occlusion make local predictions globally irreconcilable, yielding an *unsupervised* dynamic-region detector—reaching D²UST3R’s static/dynamic split from topology rather than from an optical-flow threshold. (iii) *The sheaf strictly refines the graph.* A bare correspondence graph records *that* two views are related; the sheaf records *how*, by carrying the transition maps as data, and its cohomology expresses global obstructions—scale drift, loop inconsistency, motion—that a flat union of pointmaps cannot represent. This answers, in the affirmative and precisely, whether a graph-structured world model is more expressive than a pointmap pile: not the graph, but the sheaf on it, is.

Relation to prior structure. The closest existing machinery is group synchronization: estimating absolute poses on graph vertices from relative measurements on edges [Singer, 2011], with certifiable solvers over SE(3) [Rosen et al., 2019] and, paired with a learned depth prior, over the similarity group [Yu and Yang, 2024]. Synchronization is the *linear, single-stalk* shadow of our formulation—the special case in which every stalk is one copy of the pose group and the only data are relative transforms—and it targets the same fusion stage. The sheaf view generalizes it in two ways that matter for dynamic scenes: stalks may carry dense per-view geometry rather than a single pose, and the cohomological obstruction is retained and localized rather than relaxed away. On the topology side, spectral sheaf theory [Hansen and Ghrist, 2019, 2021] and sheaf-based sensor fusion [Robinson, 2017] supply the operators we use; our contribution is their first instantiation, to our knowledge, on dense multi-view pointmap reconstruction, together with the dynamic-scene reading of H^1 .

Contributions.

1. We recast multi-view pointmap fusion as recovering a global section of a cellular sheaf on the view graph, with per-view geometry as stalks and the network’s predicted transforms as restriction maps (Section 2.3). The construction subsumes both DUST3R-style global alignment and SE(3)/Sim(3) synchronization as special cases.
2. We characterize globally consistent reconstructions as the kernel of the sheaf Laplacian and give a sparse diffusion solver with a convergence rate controlled by its spectral gap, providing a principled, analyzable replacement for the global-alignment optimizer.
3. We show that the first cohomology localizes the obstruction to consistency, and that in dynamic scenes its mass concentrates on moving and occluded regions, yielding an unsupervised dynamic-region detector that requires no flow estimator.
4. We analyze the formulation’s two honest failure modes—the non-abelian curvature of pose-valued stalks, handled by a connection (tangent-space) linearization, and the coupling between the evolving reconstruction and the sheaf it induces—and validate the resulting method empirically.

2 Preliminaries

We fix notation for pointmaps and the fusion problem (Section 2.1), recall cellular sheaves and their spectral theory (Section 2.2), and assemble the two into the reconstruction sheaf that the method operates on (Section 2.3).

2.1 Pointmaps and the fusion problem

Let $I^1, \dots, I^N$ be images of resolution $W \times H$. A *pointmap* $X^{n,m} \in \mathbb{R}^{W \times H \times 3}$ assigns to each pixel of image n its 3D location expressed in the camera frame of image m . For a pair (n, m) the network of the DUST3R family [Wang et al., 2024, Zhang et al., 2025, Han et al., 2026] regresses $X^{n,m}$ and $X^{m,m}$ together with per-pixel confidences $C^{n,m}$, from which depth, the relative camera transform, and dense correspondence follow. Writing $\pi_K^{-1}(p, d) = d K^{-1} \tilde{p}$ for the back-projection of pixel p at depth d through intrinsics K ($\tilde{p}$ homogeneous), each view n carries a depth map D_n , a pose $g_n \in \text{SE}(3)$, and intrinsics K_n , and contributes the world-frame points

$$\chi_n(p) = g_n \cdot \pi_{K_n}^{-1}(p, D_n(p)). \quad (1)$$

The view graph. Collect the views as the vertices of a *view graph* $G = (V, E)$, placing an edge $e = (i, j) \in E$ whenever the network has produced a pairwise prediction relating views i and j (in practice, pairs selected by a confidence- or overlap-based scene graph). Each edge carries the predicted relative geometry on the overlap of the two views: a relative transform g_e , a relative scale σ_e , and the correspondence field that identifies which pixels of i and j view the same scene point.

Global alignment. The standard fusion objective reconciles the pairwise predictions by minimizing a confidence-weighted pairwise disagreement,

$$\mathcal{L}_{\text{align}} = \sum_{e=(i,j) \in E} \sum_{n \in \{i,j\}} \sum_p \nu_n(p) \left\| \chi_n(p) - g_e \left(\sigma_e X_n^{(e)}(p) \right) \right\|, \quad (2)$$

over the per-view variables $\{D_n, g_n, K_n\}$ and per-edge variables $\{g_e, \sigma_e\}$, solved by gradient descent [Wang et al., 2024]. Two features of (2) motivate what follows. First, it couples the variables *only* through the shared correspondence encoded on edges—it is a sum of edge-local agreement terms—so its structure is that of a function on a graph, not on an unstructured point cloud. Second, it presumes a consistent rigid global geometry exists and seeks it by descent; it has no notion of, and returns no certificate for, the case where consistency is genuinely obstructed, which is the generic situation under object motion. The reformulation below keeps the edge-local structure of (2) but replaces “descend on the residual” with “identify the residual’s consistent part and measure its obstruction.”

2.2 Cellular sheaves and their Laplacian

We recall the minimal sheaf-theoretic vocabulary; see Curry [2014] and Hansen and Ghrist [2019] for full treatments.

Definition 1 (Cellular sheaf). A *cellular sheaf* $\mathcal{F}$ of finite-dimensional real vector spaces on a graph $G = (V, E)$ assigns

- to each vertex $v \in V$ a *stalk* $\mathcal{F}(v)$, a vector space of data local to v ;
- to each edge $e \in E$ a stalk $\mathcal{F}(e)$, the data shared on e ;
- to each incident pair $v \leq e$ a linear *restriction map* $\mathcal{F}_{v \leq e} : \mathcal{F}(v) \rightarrow \mathcal{F}(e)$.

The 0-cochains $C^0(G; \mathcal{F}) = \bigoplus_{v \in V} \mathcal{F}(v)$ collect a choice of local data at every vertex; the 1-cochains $C^1(G; \mathcal{F}) = \bigoplus_{e \in E} \mathcal{F}(e)$ collect data on every edge. Fixing an orientation on each edge $e = (i, j)$, the *coboundary* $\delta : C^0 \rightarrow C^1$ compares the two endpoints through their restriction maps,

$$(\delta x)_e = \mathcal{F}_{j \leq e} x_j - \mathcal{F}_{i \leq e} x_i, \quad e = (i, j). \quad (3)$$

Definition 2 (Global sections). A 0-cochain $x \in C^0(G; \mathcal{F})$ is a *global section* if $\delta x = 0$, i.e. if the local data agree on every edge after restriction. The space of global sections is $\Gamma(G; \mathcal{F}) = \ker \delta = H^0(G; \mathcal{F})$.

Equipping every stalk with an inner product makes δ a map of inner-product spaces with adjoint $\delta^\top$. The (*degree-0*) *sheaf Laplacian* is

$$\mathbf{L}_{\mathcal{F}} = \delta^\top \delta : C^0(G; \mathcal{F}) \rightarrow C^0(G; \mathcal{F}), \quad (4)$$

a sparse, symmetric, positive-semidefinite operator whose (i, j) block is supported exactly on the edges of G , so that $\mathbf{L}_{\mathcal{F}}$ inherits the sparsity of the view graph. When every stalk is $\mathbb{R}$ and every restriction map is the identity, $\mathbf{L}_{\mathcal{F}}$ is the ordinary graph Laplacian; the sheaf Laplacian is its data-carrying refinement. We use two standard facts.

Lemma 1 (Sections are the kernel; obstruction is cohomology [Hansen and Ghrist, 2019]). *For a cellular sheaf $\mathcal{F}$ on a graph,*

$$\ker \mathbf{L}_{\mathcal{F}} = \ker \delta = \Gamma(G; \mathcal{F}) = H^0(G; \mathcal{F}),$$

and the first cohomology $H^1(G; \mathcal{F}) = C^1 / \text{im } \delta \cong \ker \delta^\top$ is isomorphic to the space of harmonic 1-cochains. In particular $H^1 = 0$ if and only if every edge residual can be explained by a global section.

Lemma 1 is the structural backbone of the method: recovering a consistent reconstruction means projecting onto $\ker \mathbf{L}_{\mathcal{F}}$, which the heat flow $\dot{x} = -\mathbf{L}_{\mathcal{F}} x$ achieves with a rate set by the smallest nonzero eigenvalue of $\mathbf{L}_{\mathcal{F}}$; and the residual that no global section can absorb is the harmonic representative of $H^1(G; \mathcal{F})$, a quantity supported on edges and therefore *localizable*.

2.3 The reconstruction sheaf

We now place the fusion problem of (2) on the sheaf of Section 2.2. Fix the view graph $G = (V, E)$ of Section 2.1.

Stalks. The stalk $\mathcal{F}(v)$ over a view holds that view’s contribution to the world geometry. In the simplest *coordinate* model, $\mathcal{F}(v)$ is the space of world-frame point assignments χ_v of (1) for the pixels of view v ; in the *pose* model used for analysis, $\mathcal{F}(v)$ is the tangent space at the current estimate of view v ’s frame and scale, i.e. a copy of the Lie algebra $\mathfrak{se}(3) \oplus \mathbb{R}$ (the generator of $\text{Sim}(3)$). The edge stalk $\mathcal{F}(e)$ over $e = (i, j)$ is the space of geometry on the shared overlap of views i and j , into which both endpoints restrict.

Restriction maps. The restriction map $\mathcal{F}_i \lhd_e$ takes view i ’s local geometry to its prediction on the overlap of e ; concretely it applies the back-projection (1) and selects the pixels of i that the edge’s correspondence field matches to j . The two restriction maps of an edge are built so that $(\delta x)_e = 0$ encodes *exactly* the per-edge agreement term inside (2): a global section is a set of per-view geometries whose overlaps coincide after the predicted transforms, which is the definition of a globally consistent reconstruction. Thus $\mathcal{L}_{\text{align}}$ in (2) is, up to the confidence weights $\nu_n(p)$ folded into the stalk inner products, the Dirichlet energy $\frac{1}{2} \langle x, \mathbf{L}_{\mathcal{F}} x \rangle = \frac{1}{2} \|\delta x\|^2$ of the reconstruction sheaf.

Proposition 1 (Fusion as a kernel problem; informal). *With the reconstruction sheaf above, the minimizers of the global alignment energy (2) are the 0-cochains nearest to $\ker \mathbf{L}_{\mathcal{F}} = \Gamma(G; \mathcal{F})$ in the stalk inner product. When a consistent rigid reconstruction exists it is a global section and the energy attains zero; when object motion or loop inconsistency obstructs consistency, the irreducible residual is the harmonic representative of $H^1(G; \mathcal{F})$.*

Proposition 1 reframes fusion from descent on a residual to projection onto a kernel, and it identifies the leftover—the part of the data that *cannot* be made consistent—with a cohomology class rather than with optimizer noise. The dynamic-scene method developed in the sequel reads this class off the harmonic 1-cochains and attributes its support to moving and occluded regions.

Remark 1 (Two honest subtleties). First, poses live in $\text{Sim}(3)$, which is non-abelian, so the linear coboundary (3) is exact only on the tangent stalks of the pose model; on the group itself the correct object is a sheaf with a *connection*, whose curvature around a cycle is the synchronization obstruction [Singer, 2011, Rosen et al., 2019]. We work in the tangent linearization and re-linearize as the estimate updates, a sheaf-valued Gauss–Newton iteration. Second, the restriction maps depend on the correspondence field, which depends on the very geometry we solve for, so the sheaf co-evolves with its section; we treat the network’s pairwise predictions as fixed during each solve and refresh them between outer iterations. Both subtleties are analyzed in Section [TODO: method].

3 A Proof-of-Concept Experiment

The viewpoint of Section 2.3 rests on one empirical claim that no prior theory supplies: that the harmonic representative of $H^1(G; \mathcal{F})$ concentrates on moving regions rather than smearing into the static surround or being swamped by loop-closure and drift modes (Contribution 3). Before committing to the non-abelian machinery of Remark 1, we test this claim in the cleanest setting that can falsify it: a controlled synthetic scene with a single known rigid moving object, on which we build the linearized reconstruction sheaf and compute the harmonic 1-cochain numerically. The experiment uses no learned network and no training; it isolates *the sheaf machinery* from the quality of any particular pairwise predictor, and runs in seconds on CPU.

3.1 Setup

We place $N = 14$ cameras on a partial orbit viewing a scene of 90 static background points and one rigid object of 24 points that translates and rotates between frames. The view graph G is the temporal chain augmented with a few

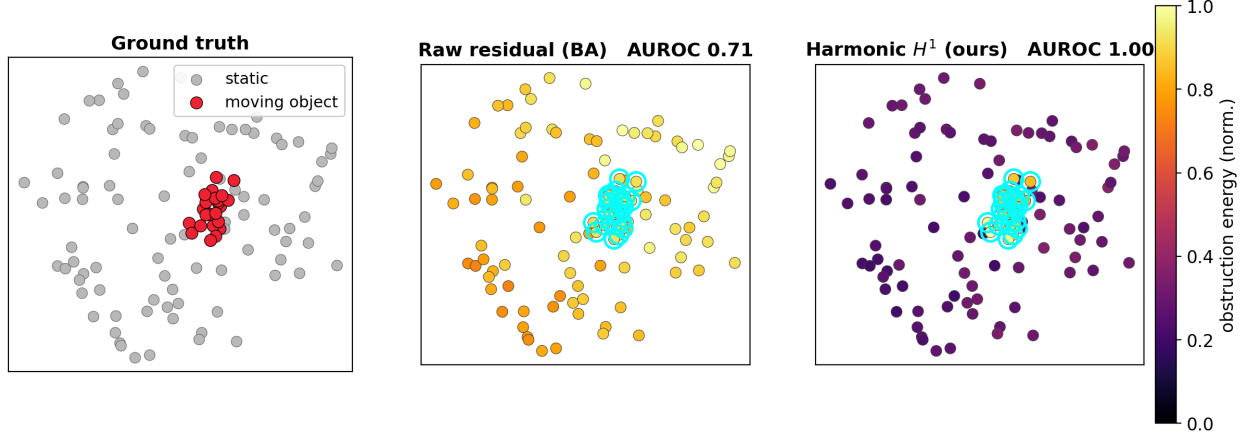


Figure 1: **The harmonic H^1 representative localizes on the moving object where the raw residual cannot.** Per-point energy on a scene with accumulating camera drift (0.30 rad/step), one projection; cyan rings mark the ground-truth moving object. *Left*: ground truth. *Middle*: the raw residual energy R_k that bundle adjustment minimizes—drift inflates static points too, so its brightest points are static and the object is not separable (AUROC 0.71). *Right*: the harmonic energy E_k , which removes everything a pose correction can explain and lights up exactly the moving object (AUROC 1.00). Energies are min–max normalized per panel.

long-range loop-closure edges, so that cycles—and hence a place for loop-inconsistency modes to live—are present. We instantiate the reconstruction sheaf in the pose model: each vertex stalk is the $\text{Sim}(3)$ tangent $\mathfrak{se}(3) \oplus \mathbb{R}$, and the edge stalk over $e = (i, j)$ is decomposed *per co-visible point*, one copy of $\mathbb{R}^3$ per point, so that a 1-cochain carries a residual for each scene point on each edge and H^1 can localize spatially rather than merely per view. For point k on edge (i, j) , each endpoint pushes its observation through the current pose estimate to a world point w_k^i , giving the base residual $r_{ij,k}^0 = w_k^i - w_k^j \in \mathbb{R}^3$; the coboundary acts by the left-perturbation Jacobian $B_{i,k} = [I \quad -[w_k^i]_\times \quad w_k^i]$, and the sheaf Laplacian is $\mathbf{L}_{\mathcal{F}} = \delta^\top \delta$, which is precisely the bundle-adjustment information matrix. The harmonic representative is the part of the residual that no pose correction can remove, $h = r^0 + \delta \xi^*$ with $\xi^* = \arg \min_\xi \|r^0 + \delta \xi\|^2$, obtained by a sparse least-squares solve (projection onto $\ker \delta^\top$, cf. Lemma 1). We compare the per-point harmonic energy $E_k = \sum_{e \ni k} \|h_{e,k}\|^2$ against the per-point raw residual energy $R_k = \sum_{e \ni k} \|r_{e,k}^0\|^2$ —the quantity bundle adjustment drives to zero—and score how well each separates the (known) moving points from the static ones by AUROC.

3.2 The obstruction localizes, and survives drift

Figure 1 shows the central result. With ground-truth poses both E_k and R_k trivially isolate the object; the interesting case is *accumulating odometry drift*, under which the static background also disagrees across views. The raw residual then conflates that disagreement with object motion and fails to localize (AUROC 0.71, and its brightest points are static), whereas the harmonic projection first subtracts off every pose-explainable disagreement and retains only the irreducible part, which sits squarely on the moving object (AUROC 1.00). This is exactly the distinction Proposition 1 draws: bundle adjustment treats the residual as noise to be minimized and discards the leftover, while the cohomological reading keeps it as a class and reads it as scene dynamics. Figure 2 quantifies the robustness across a sweep of drift magnitudes: the harmonic AUROC is pinned at 1.00 with zero variance while the raw AUROC decays from 1.00 toward chance, with a variance band that crosses below 0.5. The harmonic signal is, empirically, invariant to the camera drift that defeats the raw residual.

3.3 Stress-testing the boundaries

The clean result of Section 3.2 also exposes three structural vulnerabilities of the estimator: (i) the global scale gauge of the $\text{Sim}(3)$ coordinate model, (ii) a reliance on static-scene dominance to pin the remaining gauge, and (iii) a coboundary that is only first-order in the pose corrections. Each is a property of the estimator, not of any pairwise predictor, so each can be isolated and tested where ground truth is known and a solve costs seconds—before incurring the confounds of real data. We do so, applying a mitigation wherever one exists, and then validate the combined system on a scene that stacks all three stressors (Figure 3).

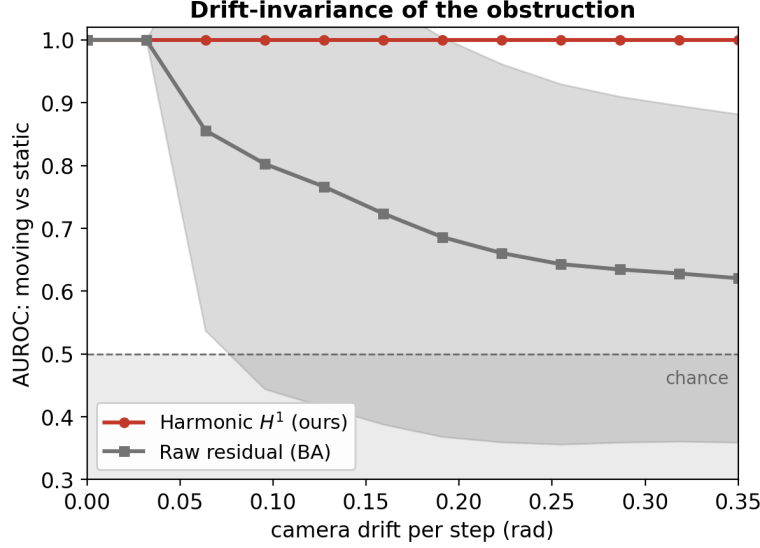


Figure 2: **The cohomological signal is invariant to camera drift; the raw residual is not.** Localization AUROC (moving vs. static) versus accumulating per-step camera drift, mean $\pm$ s.d. over 6 seeds. The harmonic energy stays at 1.00 because drift is, by construction, in the image of the coboundary and is projected away; the raw residual decays from 1.00 toward chance as drift grows. The two coincide only at zero drift—the regime the dynamic-scene problem is not about.

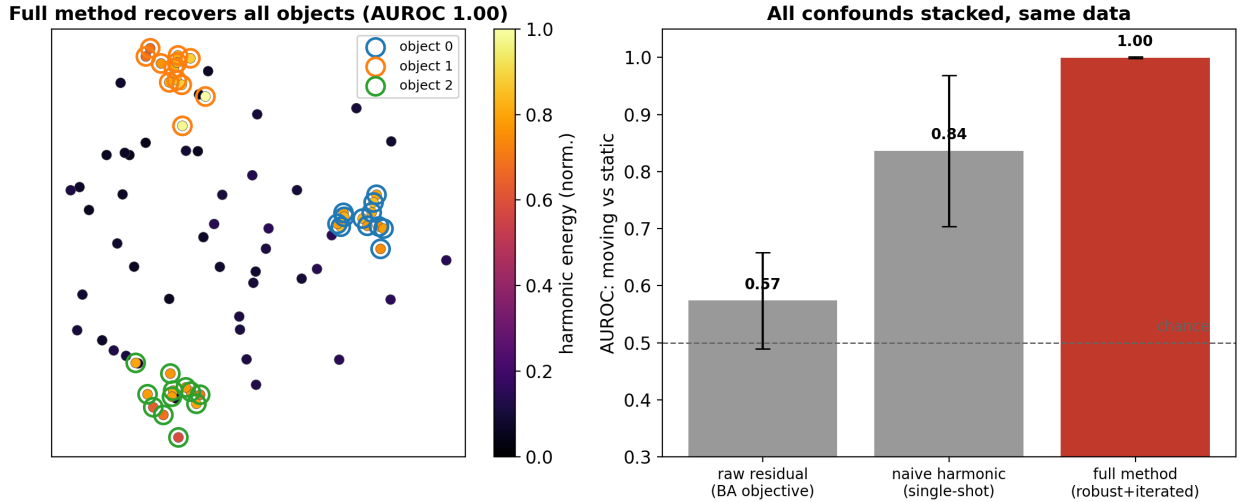


Figure 3: **All boundaries stressed at once.** A scene with three moving objects, static fraction 0.56, accumulating camera drift (0.45 rad/step), and per-view $\text{Sim}(3)$ scale ambiguity. *Left*: harmonic energy from the full solver, concentrated on all three objects (colored rings; AUROC 1.00). *Right*: localization AUROC on the same data under three readings (mean $\pm$ s.d., 6 seeds): the bundle-adjustment residual is at chance, a naive single solve is degraded and high-variance, and the full robust, iterated, scale-anchored method is near-perfect.

Scale gauge (closed). The coordinate residual $w_k^i - w_k^j$ is not scale-invariant, so the uniform log-scale direction is a coboundary that contracts the reconstruction to a point and annihilates *every* residual—the obstruction included—driving the AUROC to zero. A single linear constraint pinning the mean log-scale removes it. With the gauge anchored, localization is exactly invariant to a random global similarity (AUROC unchanged to machine precision; harmonic energy rescaling as s_g^2), and per-view scale drift is absorbed as consistent (harmonic AUROC 1.00 against a raw residual at chance). This is the metric-scale gauge of $\text{Sim}(3)$ surfacing in the spectral picture, and belongs beside Remark 1.

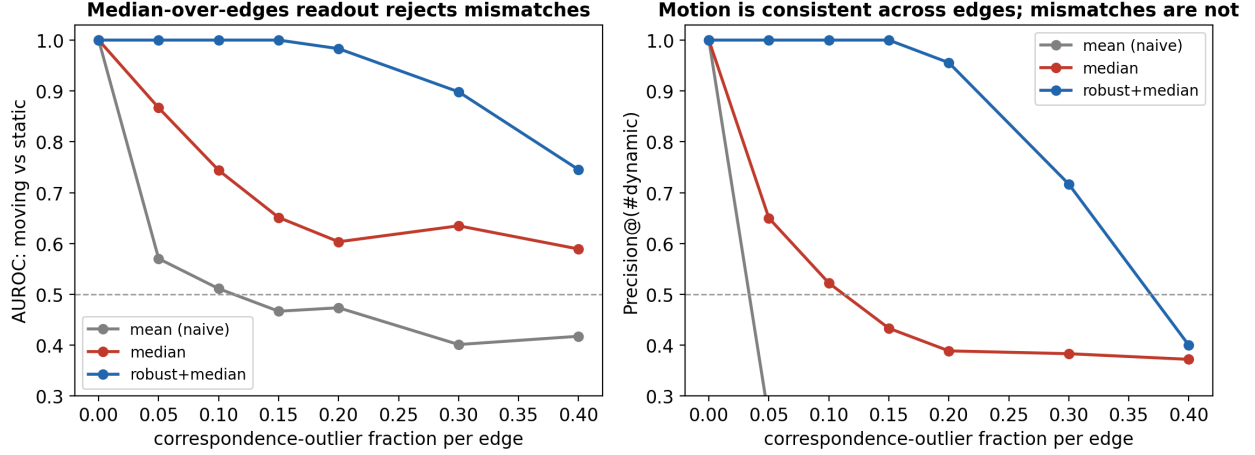


Figure 4: **Correspondence outliers.** Localization AUROC (left) and precision at the number of moving points (right) versus the per-edge mismatch rate (mean over 5 seeds). A summed energy readout collapses, because sporadic mismatches accumulate; a *median-over-edges* readout exploits that genuine motion is inconsistent on *every* edge while a mismatch is not, and with the robust solve localization is preserved to a 15–20% mismatch rate.

Linearization (controlled by iteration). Because object motion enters only through the exactly-computed residual r^0 while the linearization is in the pose corrections ξ , the “large motion” concern attaches to pose-error magnitude, not object speed. A single linearization already localizes up to large drift and fails only at the extreme; re-linearizing—the sheaf-valued Gauss–Newton of Remark 1—removes even that, holding AUROC 1.00 to a drift of 1.0 rad/step (where the single solve has collapsed to chance) and converging in two iterations. The projection suppresses only pose-explainable disagreement, so measurement noise, which is not, sets the ultimate floor.

Static dominance (sharpened, then extended). The static background pins the remaining gauge, so the precondition is not a bound on the total moving fraction but on the largest single coherent group: several small objects are each recovered even when together they exceed half the scene, since none rivals the static set. For one large object the least-squares solve degrades as its share grows, but a robust (IRLS) reweighting lets the static majority pin the gauge, holding localization down to a static fraction of ≈ 0.55 —the symmetry limit at which scene and object become indistinguishable. The separation bound a localization theorem must supply is thus governed by the dominance of the static group, not by the raw moving fraction.

3.4 Toward real data: realism stress-tests

The tests above assume exact correspondences and i.i.d. noise; real pointmap predictions violate both. We probe the three resulting gaps while ground truth is still available, and report what the real pipeline must carry as a result.

Correspondence outliers. A mismatch at a static point yields a large edge residual that is not motion. Such mismatches are *sporadic*—a point is mis-matched on only a few edges—whereas genuine motion is *consistent*, inconsistent on every edge a moving point touches. A per-point energy that *sums* over edges therefore accumulates the sporadic hits and collapses (AUROC 0.57 at a 5% outlier rate), but reading the energy as a *median* over a point’s edges, together with the robust solve, rejects them: AUROC ≈ 1.0 to a 15% rate and 0.98 at 20% (Figure 4). The residual false positives are spatially scattered rather than coherent, which a within-image smoothness prior would suppress.

Structured prediction error. Network error is spatially correlated and view-dependent, concentrating at depth and occlusion boundaries. A coherent, view-dependent error on a *static* region is—unlike i.i.d. noise—cohomologically indistinguishable from motion: it is an irreducible obstruction that neither the harmonic projection nor the robust solve removes. Only downweighting by predicted *confidence*—the per-pixel reliability the network already emits, i.e. the weights $\nu_n(p)$ of (2)—suppresses it. Confidence is thus a required input on real data, not an optional refinement.

Hyperparameters. Localization is insensitive to the solver’s settings: AUROC stays at 1.00 as the Huber constant, the IRLS and Gauss–Newton iteration counts, and the gauge-anchor weight are each varied over more than a decade.

3.5 Scope

This is a synthetic, in-the-clean test: the residuals are computed from ground-truth geometry rather than from a learned predictor, which is what lets it isolate the sheaf machinery from prediction quality. It establishes that *if* the pairwise predictions are clean enough, the harmonic H^1 provably localizes scene dynamics and does so with an invariance to pose drift that the raw residual lacks; the realism stress-tests (Section 3.4) further identify what a real pipeline must carry—a robust median-over-edges readout for correspondence outliers and per-pixel confidence weighting for structured error. It does not yet establish that a real pointmap network’s residuals meet that bar, nor does it treat occlusion—which is missing data rather than inconsistency and warrants its own test. A formal separation bound for the static-dominance precondition, now characterized and mitigated empirically (Section 3.3), likewise remains open. Validating the construction on D²USt3R predictions for a real dynamic clip is the immediate next step.

4 Real-Data Study

We evaluate the harmonic- H^1 detector on real pointmap predictions and compare it against optical-flow motion segmentation. All networks are used frozen; no training or fine-tuning is performed. The findings are mixed: the synthetic behaviour reproduces qualitatively (the harmonic signal localizes moving regions and is invariant to pose error), but as a dynamic-region detector the method does not exceed an optical-flow baseline on the data tested. We report both outcomes.

4.1 Setup

Backbones are from the DUST3R family: DUST3R [Wang et al., 2024] (rigid), MonST3R [Zhang et al., 2025] and D²USt3R [Han et al., 2026] (fine-tuned on dynamic data). Cross-view correspondence is obtained from MAST3R [Leroy et al., 2024] feature matching, an appearance-based source that uses no optical flow. Per clip we run a backbone once, take its per-view depth and poses from global alignment, form the per-correspondence world-frame residual $d_k = g_i X_i[p_i] - g_j X_j[p_j]$, compute the harmonic representative, and aggregate per-correspondence energy to pixels by a median over incident edges; scores are read only at matched pixels. Clips are five frames with a temporal chain and two loop-closure edges.

Data is the MPI-Sintel training split [Butler et al., 2012] (the only split with public ground truth). A pixel is dynamic when its observed flow departs from the camera-induced flow by more than $\tau = 3$ px, computed from ground-truth depth, pose, and flow, with occluded pixels excluded. The metric is AUROC (moving vs. static) over scored pixels.

Training-data provenance. The four networks (DUST3R, MonST3R, D²USt3R, MAST3R) are not trained on Sintel and are zero-shot on this data. The optical-flow estimator used by the flow baseline, RAFT [Teed and Deng, 2020], has a Sintel-trained variant; to avoid train–test overlap we use a RAFT model trained only on FlyingChairs and FlyingThings. The gap between the two RAFT variants on these scenes is ≤ 0.02 AUROC.

4.2 Harmonic energy vs. the alignment residual

We first compare the harmonic energy against the raw residual $\|d_k\|^2$ (the quantity global alignment minimizes) as camera poses are perturbed. Over 19 Sintel scenes the harmonic AUROC is unchanged by pose drift (0.654 at the estimated poses, 0.655 under added drift), while the raw-residual AUROC falls from 0.539 to 0.441. Localization quality depends on how much of the scene moves: across scenes AUROC is negatively correlated with the dynamic fraction (Pearson $r = -0.48$), with mean AUROC 0.72 ± 0.11 on static-dominant scenes (dynamic fraction < 0.35) and 0.47 ± 0.22 when moving content dominates. This reproduces, on real data, the static-dominance precondition identified synthetically (Section 3.3).

4.3 Comparison with optical-flow motion segmentation

The dynamic-region mask released with MonST3R and D²USt3R is the thresholded flow residual $\|\text{ego-flow} - \text{RAFT flow}\|$, where the ego-flow is computed from the backbone’s depth and poses. The comparison thus reduces to fixing the flow method and varying the geometry that feeds the ego-flow. Table 1 reports both detectors on each backbone’s geometry (six static-dominant scenes, unit stride).

The harmonic detector is strongest on the rigid backbone (0.70) and weaker on the dynamic-trained backbones, which partially remove the cross-frame inconsistency it measures. The flow detector is strongest on the most accurate

geometry	harmonic H^1	flow residual
DUS3R (rigid)	0.70	0.50
MonST3R (dynamic-trained)	0.53	0.91
D ² US3R (dynamic-trained)	0.55	0.62

Table 1: Dynamic-region AUROC by backbone geometry (mean over six static-dominant Sintel scenes, unit stride; bias-free RAFT). The harmonic detector is best on the rigid backbone; the flow detector is best on the most accurate dynamic-trained geometry.

injected drift (deg/frame)	harmonic	flow-naive	flow-refined
0 (estimated poses)	0.70	0.91	0.93
2	0.69	0.67	0.93
8	0.70	0.59	0.93
16	0.71	0.51	0.93

Table 2: Dynamic-region AUROC under injected camera drift (mean over six static-dominant scenes; each method on its best backbone). The harmonic detector is drift-invariant, but so is flow-refined, which stays above it throughout.

dynamic-trained geometry (MonST3R, 0.91). Comparing each method at its best backbone, the flow detector exceeds the harmonic detector (0.91 vs. 0.70).

Behaviour under pose error. Because the ego-flow depends on camera poses, we test whether the harmonic detector’s pose-invariance yields an advantage as poses degrade. We inject accumulating camera drift and report three curves (Table 2): the harmonic detector; the flow detector using the drifted poses directly (flow-naive); and the flow detector after re-estimating poses on static correspondences with the same robust solver (flow-refined). The harmonic detector is invariant to drift and flow-naive degrades, as expected; however flow-refined is also invariant to drift and remains above the harmonic detector at every level. Re-estimating poses from static structure is inexpensive, so pose-invariance does not separate the harmonic detector from a flow detector allowed to refine its poses. The estimated-pose operating point ($\approx 3.9^\circ/\text{frame}$ from ground truth) already lies in the regime where the flow detector is at least as accurate. On the five dynamic-heavy scenes, flow-refined is at least as accurate as the harmonic detector on all of them.

Consistency check with D²US3R. We also tested whether building the sheaf on D²US3R’s motion-aligned pointmaps suppresses H^1 , since that model warps dynamic pixels to be consistent across time. On the scenes tested this did not occur: the harmonic energy under D²US3R geometry was comparable to that under DUS3R. We confirmed that d_k responds strongly to the geometry change, so the operator is not insensitive to it; the alignment D²US3R applies does not coincide with the world-frame cross-time consistency the operator measures.

4.4 Robust fusion vs. global alignment

A second use of the construction is as a robust replacement for global alignment itself, rather than as a detector. The alignment objective is the sheaf Dirichlet energy, and the harmonic-projection / IRLS robustness downweights cross-frame-inconsistent pixels, which could in principle keep moving content from contaminating the static reconstruction. We compare standard DUS3R global alignment (a) against the same optimization with its per-pixel weights reweighted by a Huber factor on its own residual (b), and measure depth error against ground truth, separated into static, dynamic, and all pixels. Depth is aligned to ground truth by an affine (scale and shift) fit before computing AbsRel. Table 3 reports the mean over six static-dominant and three dynamic-heavy scenes.

On these clips the static reconstruction is determined by the static majority and is not strongly contaminated by the moving minority, so there is little for the robust weighting to recover; in addition, as noted in Section 3.4, residual-based downweighting cannot separate moving content from structured static prediction error. We therefore do not find a reconstruction-quality advantage for robust fusion over global alignment on this data. (A camera-pose error metric was attempted but is omitted owing to an unresolved convention mismatch between the estimated and ground-truth cameras; the near-identical depth of (a) and (b) indicates the poses also differ little.)

scene group	region	standard GA	robust
static-dominant (6)	static	0.547	0.549
	dynamic	0.313	0.314
	all	0.739	0.738
dynamic-heavy (3)	static	1.063	1.067
	dynamic	0.276	0.276
	all	2.290	2.266

Table 3: Depth AbsRel vs. ground truth (lower is better), standard global alignment vs. robust reweighting. Robust reweighting changes the solution marginally and does not reduce static-region error; on static pixels it is slightly worse, and the effect does not grow on the dynamic-heavy scenes where contamination would be largest.

4.5 Summary

On the data tested, the harmonic- H^1 detector localizes moving regions when the scene is static-dominant, and its signal is invariant to camera-pose error. As a dynamic-region detector it does not exceed optical-flow motion segmentation that uses an accurate backbone and standard pose re-estimation, on clean geometry or under pose drift, on the Sintel scenes considered; its accuracy is further limited when moving content is not a minority. Used instead as a robust fusion step, it does not improve static-region reconstruction over standard global alignment (Section 4.4). These observations qualify the unsupervised-detector claim of the introduction: on this evidence the contribution is the cohomological formulation of fusion and its synthetic characterization, rather than a real-data method that improves on flow-based segmentation or on global alignment.

5 Loop-closure consistency on the pose sheaf

The dense per-point model of Sections 2.3–4 repeatedly meets the same obstacle: a structured, view-dependent prediction error on a static region is cohomologically indistinguishable from object motion (Section 3.4). We therefore ask whether the same construction is more useful on a coarser sheaf, where that confound is averaged away before it can become an obstruction.

5.1 The pose sheaf

Keep the view graph $G = (V, E)$, but place the sheaf on the *pose* model. Per edge $e = (i, j)$ the backbone yields a measured relative transform $M_{ij} \in \text{SE}(3)$ (recovered below from the pairwise pointmaps). Take stalks $\mathcal{F}(v) = \mathcal{F}(e) = \mathfrak{se}(3)$ and restriction maps given by the connection, $\mathcal{F}_i \leq e = \text{Ad}_{M_{ij}^{-1}}$ and $\mathcal{F}_j \leq e = I$, so that with a right perturbation $g_k \leftarrow g_k \exp(x_k)$ the coboundary linearizes to

$$(\delta x)_e = x_j - \text{Ad}_{M_{ij}^{-1}} x_i, \quad r_e^0 = \log(M_{ij}^{-1} g_i^{-1} g_j). \quad (5)$$

Then $\mathbf{L}_{\mathcal{F}} = \delta^\top \delta$ is exactly the pose-graph information matrix, and the harmonic representative of H^1 is the part of the edge residual that no choice of frames can remove: the irreducible loop-closure obstruction, whose projection onto a cycle is its linearized log-holonomy. The obstruction space has dimension $\dim \mathfrak{g} \cdot b_1(G)$ (tens of independent cycles), not the $\sim 10^6$ of pixel space. For real data we use the rotation-only $\text{SO}(3)$ sub-sheaf (restriction $R_{ij}^\top$), which is invariant to the per-pair scale ambiguity of pointmap-derived poses.

5.2 Synthetic validation

On a closed camera orbit (10 views, ring plus chords) with measurements $M_{ij} = g_i^{-1} g_j$ the operator behaves as predicted (Table 4). Consistent measurements give zero harmonic energy to machine precision; a single edge corrupted by a rotation θ is localized by per-edge harmonic leverage $\|h_e\|^2$ with AUROC > 0.95 once θ exceeds the measurement noise; and the obstruction is invariant to camera-pose drift, provided the linearization is iterated to convergence (a single linearization is only first-order invariant, as r^0 is a logarithm of a product).

5.3 Real data: scene-graph pruning on TUM RGB-D

We test the practical use the operator suggests — pruning the DUST3R scene graph before global alignment — on TUM RGB-D fr1/room [Sturm et al., 2012], a handheld loop with motion-capture ground-truth poses; we use 30

check	quantity	result
consistent measurements	$\max \ h_e\ $	1.9×10^{-16}
planted bad edge ($\theta \geq 0.05$)	bad-edge AUROC	0.96–0.97
drift invariance	harmonic energy ratio	median 1.000

Table 4: Pose-sheaf operator on a synthetic closed orbit. The obstruction is zero when measurements are consistent, localizes a planted bad edge, and is invariant to the absolute-frame estimate (camera-pose drift).

selection	$K=5$	$K=10$	$K=20$	$K=40$
confidence	28.3	26.5	30.1	30.2
sheaf leverage	26.2	76.5	22.2	21.9
oracle (GT)	28.2	36.4	30.1	23.4
random	47.2	30.6	32.8	70.9

Table 5: Trajectory error (ATE, cm; lower is better) after global alignment on a graph of the consecutive-frame backbone plus K loop edges, by selection criterion. Backbone alone gives 59.7 cm. Confidence selection is already near the oracle; sheaf leverage does not improve on it and is unstable on this graph.

keyframes. Per pair we recover the relative pose by a confidence-weighted Sim(3) Procrustes fit between the two pixel-aligned pointmap predictions of the shared image. Against the mocap ground truth, high-overlap pairs have a median rotation error of 2° and genuine revisit edges (temporally distant, spatially near) a median of 3° , confirming both the pose convention and that DUST3R produces accurate loop closures. However, over the complete pairwise graph the relative-pose error is *dense* rather than *sparse-outlier* (negative excess kurtosis), since most pairs barely overlap, and the error is largely predicted by signals already available: its correlation with the ground-truth viewpoint change is 0.81 and with low pairwise confidence 0.49.

We keep the consecutive-frame backbone and add K loop edges selected by DUST3R confidence, by sheaf leverage (lowest harmonic mass), by an oracle (lowest ground-truth rotation error), or at random, then run the same global alignment and measure absolute trajectory error (ATE) against ground truth (Table 5). Two effects appear. First, confidence already selects near-oracle loop edges — the mean ground-truth rotation error of its top 20 picks is 1.4° , against the oracle’s 0.8° — so there is little headroom for any consistency-based criterion to improve on it. Second, sheaf leverage performs poorly here (its top 20 picks average 25.4°): the harmonic criterion presumes a predominantly consistent graph with sparse outliers, whereas the complete pairwise graph is predominantly low-overlap, so the rotation-synchronization consensus is itself unreliable and the few good edges, which disagree with it, are flagged as inconsistent.

5.4 Summary

The pose-sheaf construction is sound and validated in synthesis: H^1 is the loop-closure obstruction, localizes a planted bad edge, and is drift-invariant. It does not, however, yield a real-data advantage for scene-graph pruning, for two data-dependent reasons that also explain the negative results of Section 4: the backbone’s pairwise graph is *dense-error* rather than *sparse-outlier*, which is the regime the harmonic criterion requires, and the backbone’s own confidence already selects loop edges near-optimally, leaving no headroom. The contribution that survives our study is therefore the cohomological formulation itself and its synthetic characterization, rather than a demonstrated improvement on a frozen-backbone reconstruction or SLAM pipeline.

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